

CISCO SYSTEMS

Enterprise Lab Series

Quality of Service (QoS)

Modular QoS CLI — Production-Style Lab

CCNP Enterprise

350-401 ENCOR | 300-410 ENARSI

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2-Node Isolation Topology

HQ-Core (Traffic Generator) ↔ HQ-Edge (Policy Engine)



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1. Lab Overview

This production-style Quality of Service (QoS) lab is tailored for CCNP Enterprise examination preparation (350-401 ENCOR and 300-410 ENARSI). It isolates advanced QoS concepts using a streamlined **2-Node Topology** to maximize study efficiency and eliminate the resource overhead of a large-scale mega lab.

By focusing purely on the data plane interaction between a traffic generator and the policy engine, you will master the **Modular QoS CLI (MQC)** framework, Hierarchical QoS (HQoS), shaping, policing, and advanced congestion management.

1.1 Learning Objectives

- **MQC Framework:** Build class maps, policy maps, and service policies end-to-end
- **Trust Boundary:** Establish DSCP marking at the LAN ingress point using ACLs and NBAR2
- **LLQ / CBWFQ:** Configure Low Latency Queuing for voice and Class-Based WFQ for data
- **WRED:** Prevent TCP global synchronization using Weighted Random Early Detection
- **HQoS:** Shape a substrate WAN link to force software queue engagement
- **Policing:** Hard-drop scavenger traffic at ingress with a two-color rate limiter

1.2 Prerequisites

- Familiarity with Cisco IOS-XE CLI (CSR1000v, Catalyst 8000v, or IOL)
- Basic understanding of IP routing (CEF, static routes)
- EVE-NG or GNS3 virtual lab environment
- Two virtual nodes: one IOL/vIOS (Core), one IOS-XE (Edge)

2. Lab Topology & Setup

2.1 The 2-Node Isolation Strategy

In this optimized layout, we remove the multi-hop WAN service provider and branch nodes. **HQ-Edge** handles all ingress marking and egress scheduling internally. To trigger egress queuing on a WAN port with no real neighbor, we point a static host route to a dummy gateway out of the egress interface, forcing the router's software buffers to process outbound traffic queues under heavy load.

2.2 Device Roles

- **HQ-Core:** Lightweight IOL or vIOS node acting as a high-speed multi-application traffic generator
- **HQ-Edge:** Robust IOS-XE node (CSR1000v or Catalyst 8000v) acting as the enterprise edge — all classification, marking, shaping, scheduling, and WRED occur here

2.3 Interface & IP Addressing

Device	Interface	IP Address	Connecting To
HQ-Core	Ethernet0/0	10.1.1.2/24	HQ-Edge Gi1
HQ-Edge	GigabitEthernet1	10.1.1.1/24	HQ-Core Eth0/0
HQ-Edge	GigabitEthernet2	192.168.12.1/30	Dummy WAN (Unconnected)

3. Scenario & Traffic Profile

Your company has a **10 Mbps subrate FastE WAN link** delivered over a physical GigabitEthernet interface (Gi2) on HQ-Edge. You must build an end-to-end MQC policy to handle traffic generated from HQ-Core according to the corporate QoS profile below:

Traffic Type	Classification Method	Marking	QoS Strategy
Voice (VoIP)	UDP Ports 16384–32767	DSCP EF	LLQ (Strict Priority), 10% Bandwidth Max
Video Conf.	Source IP: 10.1.1.2	DSCP AF41	CBWFQ — 20% Bandwidth
Critical Data	Protocol: http / https	DSCP AF21	CBWFQ (30% BW) + WRED Drop Avoidance
Scavenger	Protocol: bittorrent / youtube	DSCP CS1	Ingress Policing to 128 Kbps

4. Lab Tasks

Task 1: Ingress Classification & Marking (HQ-Edge)

Theory & Intent

Establishing the Trust Boundary: We classify traffic at the closest ingress point (Gi1) where it enters from the LAN. Internal traffic arrives raw and untrusted. Applying a standard 6-bit **DSCP (Differentiated Services Code Point)** tag on the IP header immediately allows downstream egress engines to execute scheduling with minimal CPU overhead.

ACLs vs. NBAR2: Standard Access-Lists examine static L3/L4 parameters such as port ranges (UDP RTP streams) or host IPs. **Network-Based Application Recognition (NBAR2)** utilizes deep packet inspection (DPI) to identify complex application protocols like web traffic (http/https) where port numbers alone cannot distinguish essential business platforms from general browsing.

Implementation Steps

1. Configure extended ACLs on HQ-Edge to trap VoIP and video streams
2. Build ingress class maps to group the defined traffic profiles
3. Build the MARKING-INGRESS policy map to stamp DSCP tags
4. Apply the policy to the ingress side of interface GigabitEthernet1

Configuration

```
ip access-list extended ACL_VOICE_RTP
  permit udp any any range 16384 32767
!
ip access-list extended ACL_VIDEO_CONFERERENCE
  permit ip host 10.1.1.2 any
!
class-map match-any CM_VOICE
  match access-group name ACL_VOICE_RTP
class-map match-any CM_VIDEO
  match access-group name ACL_VIDEO_CONFERERENCE
class-map match-any CM_CRITICAL
  match protocol http
  match protocol https
class-map match-any CM_SCAVENGER
  match protocol bittorrent
  match protocol youtube
!
policy-map MARKING-INGRESS
  class CM_VOICE
    set dscp ef
  class CM_VIDEO
    set dscp af41
  class CM_CRITICAL
    set dscp af21
  class CM_SCAVENGER
    set dscp cs1
  class class-default
    set dscp default
!
interface GigabitEthernet1
  ip nbar protocol-discovery
  service-policy input MARKING-INGRESS
```

Task 2: Egress Queuing & Congestion Management

Theory & Intent

DSCP Optimization: At egress, we match DSCP values instead of reproducing complex ACL checks, keeping the scheduler CPU-efficient.

Low Latency Queuing (LLQ): The `priority` command establishes a strict priority queue for Voice (EF). When a packet arrives here, the scheduler halts all other queues to transmit it instantly. A percentage cap (`percent 10`) acts as a circuit breaker — if a voice sub-system floods the network, it cannot starve remaining data applications.

Class-Based Weighted Fair Queuing (CBWFQ): The `bandwidth` command guarantees a minimum operational slice of the link only during congestion. If the link is under-utilized, these classes safely burst past their assigned limits.

Weighted Random Early Detection (WRED): When a buffer fills completely, a burst of packets drops simultaneously (Tail Drop), causing every sending application to back off concurrently — a volatile cyclic crash known as **TCP Global Synchronization**. WRED (`random-detect`) acts preemptively by dropping individual packets at random as thresholds expand, forcing specific TCP windows to downscale gracefully before buffer exhaustion.

Implementation Steps

5. Build egress class maps matching the DSCP tags created in Task 1
6. Build egress child scheduling policy map named QUEUE-EGRESS
7. Allocate LLQ to Voice, CBWFQ to Video/Data, and bind DSCP-based WRED to the critical data queue

Configuration

```
class-map match-any CM_WAN_VOICE
  match dscp ef
class-map match-any CM_WAN_VIDEO
  match dscp af41
class-map match-any CM_WAN_CRITICAL
  match dscp af21
!
policy-map QUEUE-EGRESS
  class CM_WAN_VOICE
    priority percent 10
  class CM_WAN_VIDEO
    bandwidth percent 20
  class CM_WAN_CRITICAL
    bandwidth percent 30
    random-detect dscp-based
  class class-default
    fair-queue
```

Task 3: Hierarchical QoS (HQoS) — Egress Shaping

Theory & Intent

The Hardware Illusion: Interface Gi2 operates at a physical clock speed of 1 Gbps. If you attach queuing configuration directly to this physical interface, the system assumes it has 1,000 Mbps of headroom. Passing a 40 Mbps stream shows zero local bottleneck — the output buffer rings never engage, and the child policy queues sit idle

The Subrate Backlog Generation: Because your contracted carrier limit is only 10 Mbps, the provider switch drops anything past that value. To force your router to manage congestion before sending frames out, we construct a **Hierarchical QoS (HQoS)** parent policy. The parent shapes the total egress envelope to 10

Mbps, creating a regulated virtual software backlog within the router and forcing the nested QUEUE-EGRESS schedule to engage accurately.

Implementation Steps

8. Create a parent policy map named SHAPE-10MBPS-PARENT
9. Configure a class-default shaper targeting an average rate of 10,000,000 bps
10. Nest the QUEUE-EGRESS child execution profile directly inside the shaper
11. Apply the parent configuration to egress interface GigabitEthernet2
12. Write a dummy static route to force test traffic through the egress pipeline

Configuration

```
policy-map SHAPE-10MBPS-PARENT
  class class-default
    shape average 10000000
    service-policy QUEUE-EGRESS
  !
interface GigabitEthernet2
  service-policy output SHAPE-10MBPS-PARENT
  !
! Dummy route to force traffic out of Gi2 into the egress queues
ip route 8.8.8.8 255.255.255.255 GigabitEthernet2 192.168.12.2
```

Task 4: Ingress Policing — Scavenger Control

Theory & Intent

Shaping vs. Policing: Shaping queues delayed packets in buffer memory to smooth out transit peaks over time (ideal for WAN egress lines). Policing enforces a rigid, instantaneous speed ceiling — if an application exceeds that ceiling, traffic is dropped immediately on ingress.

Scavenger Strategy: Non-essential resource hogs (peer-to-peer transfers, YouTube streaming) should not consume internal router memory buffers. By deploying an ingress policer, we execute hard drops at the entrance interface (Gi1) before the packet can draw down core system resources.

Implementation Steps

13. Modify the primary MARKING-INGRESS policy map
14. Bind a single-rate, two-color rate-limiter set to 128,000 bps under the CM_SCAVENGER class

Configuration

```
policy-map MARKING-INGRESS
  class CM_SCAVENGER
    set dscp cs1
    police 128000 conform-action transmit exceed-action drop
```

5. Complete Core Node Configuration (HQ-Core)

Apply this minimal routing and interface configuration to your lightweight IOL/vIOS traffic generation node:

```
hostname HQ-Core
!
interface Ethernet0/0
  description CONNECTS_TO_HQ-EDGE_GI1
  ip address 10.1.1.2 255.255.255.0
  no shutdown
!
! Default gateway pointing to HQ-Edge to forward all outbound test traffic
ip route 0.0.0.0 0.0.0.0 10.1.1.1
```

6. Verification & Educational Analysis

6.1 ToS Decimal Reference

Standard Cisco ping does not accept DSCP names directly. Supply the decimal equivalent of the Type of Service (ToS) byte in the IP header:

Traffic Class	ToS Decimal	DSCP Name / Value
Voice (VoIP)	184	EF — Expedited Forwarding (DSCP 46)
Video Conference	136	AF41 — Assured Forwarding 4,1 (DSCP 34)
Critical Data	72	AF21 — Assured Forwarding 2,1 (DSCP 18)
Scavenger	8	CS1 — Class Selector 1 (DSCP 8)

6.2 Running the Test Streams (From HQ-Core)

```
! Stream 1: Test Priority LLQ — Voice matching
ping 8.8.8.8 size 100 repeat 1000 tos 184

! Stream 2: Flood Critical Data with large packets to trigger WRED drops
ping 8.8.8.8 size 1400 repeat 3000 tos 72 timeout 0
```

6.3 Active Policy Inspection (On HQ-Edge)

```
show policy-map interface GigabitEthernet2
```


7. Virtual Lab Environment Note

Engineering Reassurance — Expected Behavior in EVE-NG / GNS3

When running this lab inside a virtual environment, you might occasionally see ingress counters increment correctly (e.g., CM_VIDEO catching thousands of packets) while egress counters on GigabitEthernet2 remain at zero. Do not panic and do not change your configurations. In a virtual environment, virtual interfaces lack real physical line-state signaling. If an egress interface does not detect a live physical carrier, the router's data plane may bypass the outbound software queues or silently discard packets at the CEF switching layer before they reach the policy engine. If your syntax matches this guide, your structural logic is correct. In production networks where links are physically active, these exact commands trigger scheduling perfectly. Focus on mastering the configuration hierarchy and the underlying theory — that is what gets you certified.

8. Lab Challenge Integration

Alter the scheduling behavior so that during periods of intense link congestion, Video traffic (AF41) is **capped at a strict ceiling of 2 Mbps**, ensuring it cannot consume raw unallocated link headroom at the expense of standard corporate background data (class-default).

8.1 Challenge Solution

```
policy-map QUEUE-EGRESS
  class CM_WAN_VOICE
    priority percent 10
  class CM_WAN_VIDEO
    bandwidth percent 20
    police 2000000 conform-action transmit exceed-action drop
  class CM_WAN_CRITICAL
    bandwidth percent 30
    random-detect dscp-based
  class class-default
    fair-queue
```

8.2 Explanation

Adding the policer under `CM_WAN_VIDEO` enforces a **hard ceiling of 2 Mbps** on video traffic. Without this policer, CBWFQ only guarantees a minimum bandwidth of 20% during congestion; when the link is lightly loaded, the video class can burst to consume all available headroom. The added `police` statement creates a two-color meter that transmits conforming traffic and drops exceeding traffic regardless of link utilization state.

9. Quick Reference — Policy Map Hierarchy

The complete MQC policy hierarchy applied on HQ-Edge can be summarized as follows:

```
GigabitEthernet1 (Ingress)
├─ MARKING-INGRESS
│   ├── CM_VOICE → set dscp ef
│   │               police 128000 [Voice – no policing; Scavenger below]
│   ├── CM_VIDEO → set dscp af41
│   ├── CM_CRITICAL → set dscp af21
│   ├── CM_SCAVENGER → set dscp cs1 + police 128000 conform=transmit exceed=drop
│   └─ class-default → set dscp default
└─
GigabitEthernet2 (Egress)
├─ SHAPE-10MBPS-PARENT [Parent – shapes to 10 Mbps]
│   └─ QUEUE-EGRESS [Child – schedules within 10 Mbps envelope]
│       ├── CM_WAN_VOICE → priority percent 10 (LLQ strict)
│       ├── CM_WAN_VIDEO → bandwidth percent 20 (CBWFQ)
│       ├── CM_WAN_CRITICAL → bandwidth percent 30 + random-detect dscp-based
│       └─ class-default → fair-queue
```

— *End of Lab Guide* —

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